AP PRECALCULUS

Unit 1 • Worksheet 1.12B • Answer key

Worksheet 1.12B • Transformations of Functions

Answer key with worked solutions

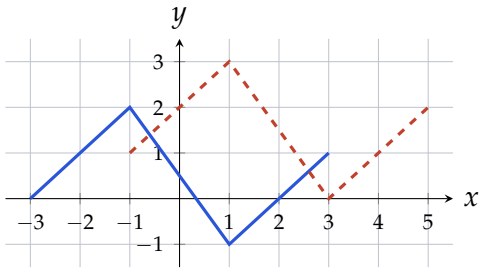
Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

The graphs of f (solid) and g (dashed) are shown. Which is g in terms of f?



(A) $f(x-2)+1$

(B) $f(x+2)+1$

(C) $f(x-2)-1$

(D) $f(x+2)-1$

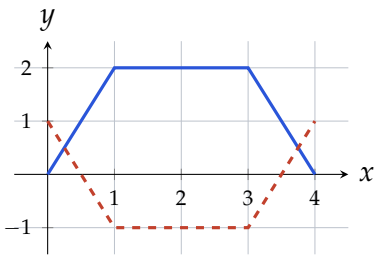
Solution

Answer: (A). Right 2, up 1.

Question 2

NO CALCULATOR

The graphs of f (solid) and g (dashed) are shown. Which is g in terms of f?



(A) $-f(x)-1$
(B) $f(-x)+1$
(C) $-f(x)+1$
(D) $-f(x+1)$

Solution
Answer: (C). Reflect over the x-axis, up 1.

Question 3 (Free response)

NO CALCULATOR

x	-2	-1	0	1	2
$f(x)$	5	2	1	2	5

Complete a table for each function, where defined: (a) $g(x) = f(x) - 3$ (b) $h(x) = 2f(x)$ (c) $k(x) = f(x+1)$ on $x = -3, \ldots, 1$.

Solution
(a) $2, -1, -2, -1, 2$. (b) $10, 4, 2, 4, 10$. (c) $k(-3) = 5, k(-2) = 2, k(-1) = 1, k(0) = 2, k(1) = 5$.